

## WEEK 3 Report

# DEEP LEARNING BASED FEATURE EXTRACTION FOR IRIS AUTHENTICATION

### Objective

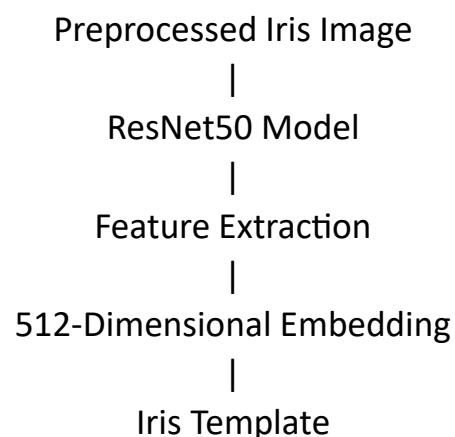
The objective of this work is to generate iris feature vectors using deep learning models. These feature vectors will be used as iris templates for secure biometric authentication in the later phases of the project.

### ResNet50 Based Feature Extraction

In this notebook, the pretrained ResNet50 model was used for iris feature extraction. The preprocessing images generated in Week 2 were used as input to the model. Each image was resized to  $224 \times 224$  pixels before feature extraction.

The output of the model was converted into a 512-dimensional feature vector. These feature vectors represent the important iris patterns of each image.

### Workflow:



## **ResNet50 Feature Extraction Results**

- 1) Successfully loaded the pretrained ResNet50 model.
- 2) Used the model as a feature extractor for iris images.
- 3) Converted each iris image into a 512-dimensional feature vector.
- 4) Generated embeddings for training, validation and test images.
- 5) Obtained an embedding output shape of (1, 512) for each image.
- 6) Saved all generated embeddings in CSV format for future processing.
- 7) Created iris templates that can be used for authentication and encryption.

## **MobileNetV2 Model Comparison**

A second deep learning model, MobileNetV2, was implemented to compare its performance with ResNet50. MobileNetV2 is a lightweight deep learning architecture designed for faster execution and lower computational cost.

## **MobileNetV2 Comparison Results**

- 1) Successfully implemented MobileNetV2 for iris feature extraction.
- 2) Generated 512-dimensional embeddings using MobileNetV2.
- 3) Processed 100 sample iris images for comparison.
- 4) MobileNetV2 contained 2,913,856 parameters.
- 5) Saved MobileNetV2 embeddings for analysis.

## **Comparison Between ResNet50 and MobileNetV2:**

| <b>Feature</b>     | <b>ResNet50</b> | <b>MobileNetV2</b> |
|--------------------|-----------------|--------------------|
| Parameter Count    | 25.6 Million    | 2.91 Million       |
| Embedding Size     | 512             | 512                |
| Memory Requirement | high            | Low                |
| Speed              | Slower          | Faster             |

## **Conclusion**

- > Successfully implemented ResNet50 and MobileNetV2 for iris feature extraction.
- > Generated 512-dimensional iris embeddings.
- > Created iris templates for training, validation, and test datasets.
- > Compared ResNet50 and MobileNetV2 based on model size and computational efficiency.
- > Found that MobileNetV2 is lighter and faster, while ResNet50 provides stronger feature extraction capability.